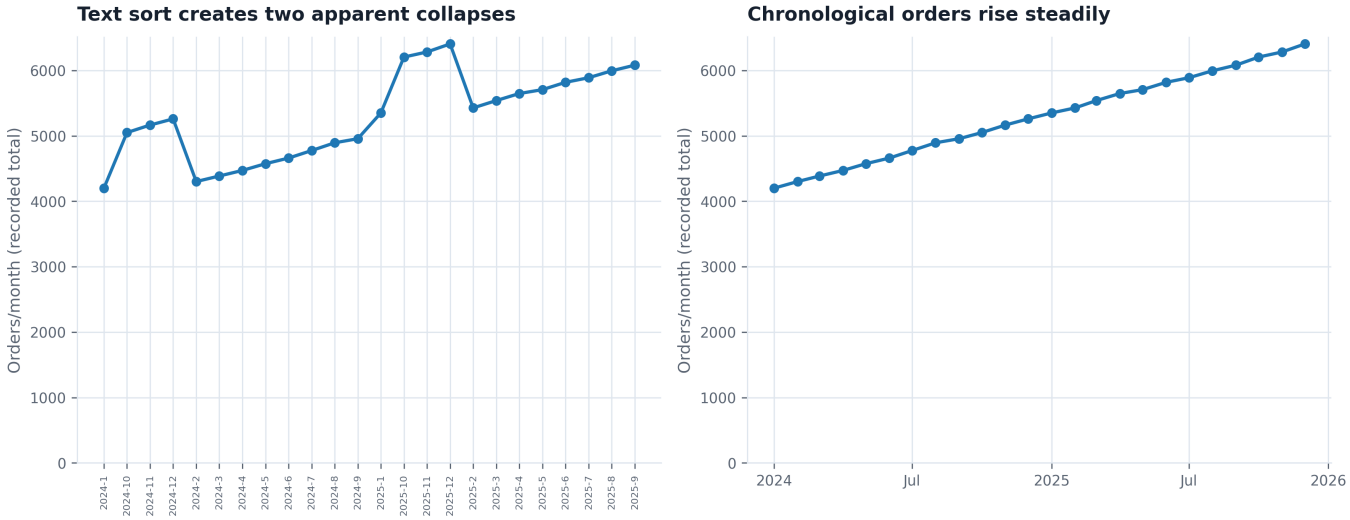
Case 1: Lexicographic dates



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

Orders appear to collapse annually.

Decoy

Not a truncated y-axis: both start at zero; branch is constant across all 24 rows.

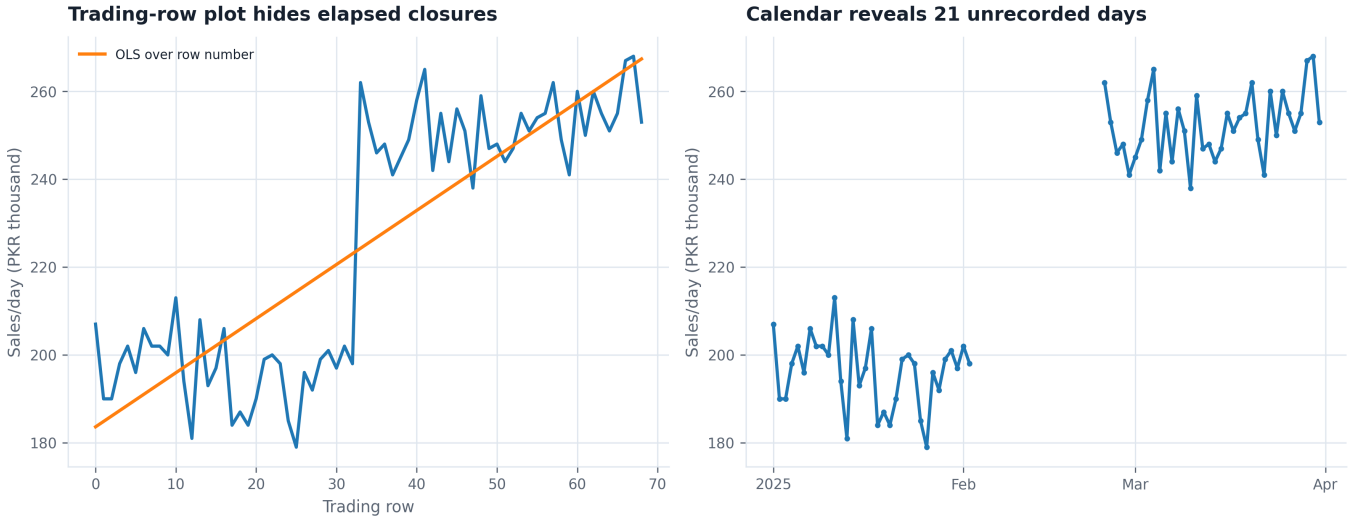
Mechanism

Text order puts October-December between January and February in each year; December to February causes drops [-958.0, -978.0]. Chronological minimum monthly change is 59; first to last 4200 to 6405.

Redesign

Parse YYYY-M explicitly, sort DatetimeIndex and use a calendar axis.

Case 2: Time compression



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

A 22% endpoint rise appears to be steady quarter-wide growth.

Decoy

The endpoint arithmetic is true: 22.22%; it does not establish a regular daily trend.

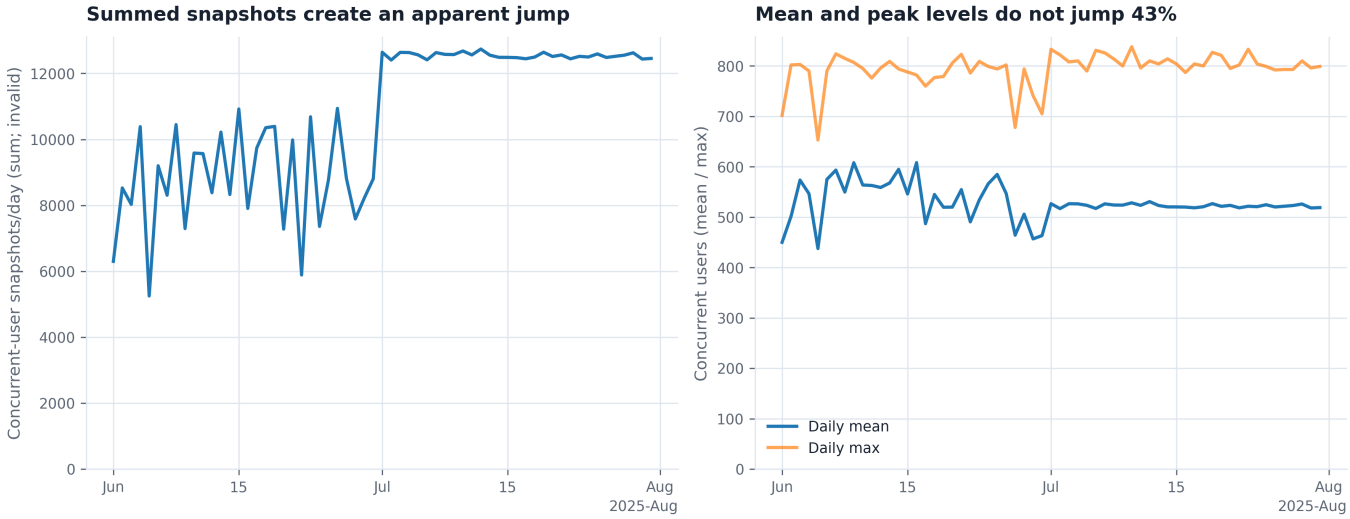
Mechanism

69 observations span 90 days; 21 absent days vanish on a row-number axis. Monthly recorded totals: {Timestamp('2025-01-01 00:00:00'): 6076.0, Timestamp('2025-02-01 00:00:00'): 1650.0, Timestamp('2025-03-01 00:00:00'): 7834.0}.

Redesign

Use dates and preserve gaps; do not impute zero without verified closure records.

Case 3: A level summed over unequal coverage



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

Peak demand seems to jump in July.

Decoy

The axis is zero-based; bytes served is a flow, but concurrent users is a stock/level.

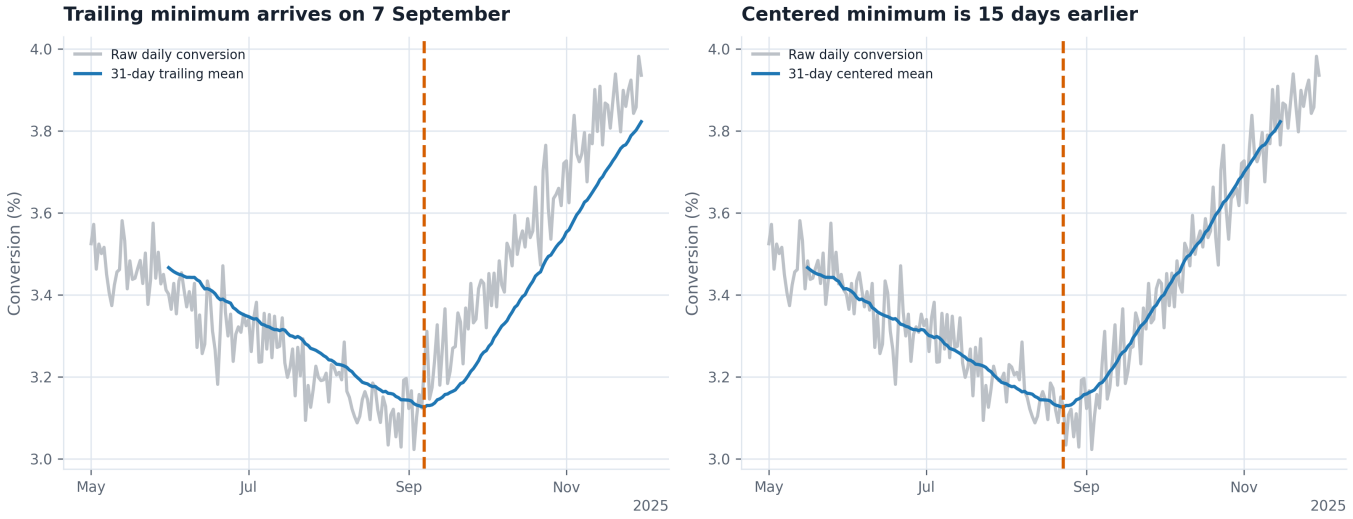
Mechanism

June/July daily statistics: {'wrong_sum': {6: 8782.433333333332, 7: 12543.129032258064}, 'mean_level': {6: 535.8873933624283, 7: 522.630376344086}, 'samples': {6: 16.433333333333334, 7: 24.0}}. Summing more snapshots scales with observation count, not simultaneous demand.

Redesign

Show daily mean and maximum level; report observation counts. Use sum only for bytes served.

Case 4: Filter phase lag



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

The checkout change appears to start working on 7 September.

Decoy

This is not evidence of a second intervention; a centered filter uses future data and cannot be deployed live.

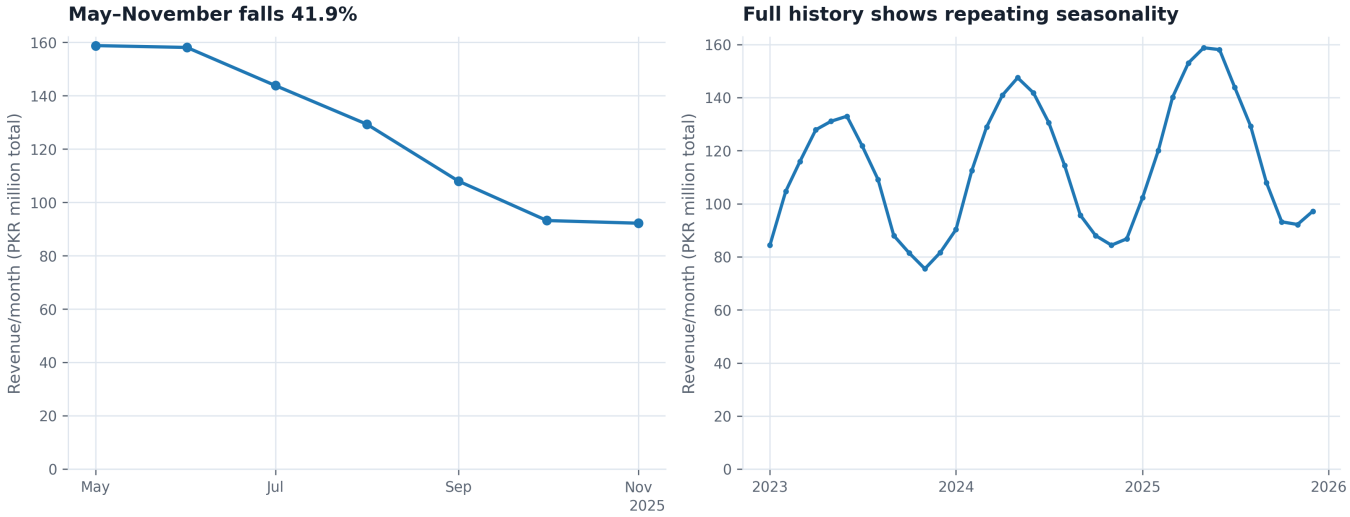
Mechanism

Trailing minimum 2025-09-07, centered minimum 2025-08-23, lag 15 days = half-window 15.

Redesign

Retrospectively show centered mean with raw data; label trailing lag on live charts. No causal attribution from timing alone.

Case 5: Seasonal window selection



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

Revenue seems to be persistently deteriorating.

Decoy

The recipe explicitly starts the axis at zero; there are 36 complete months, not a seven-month source file.

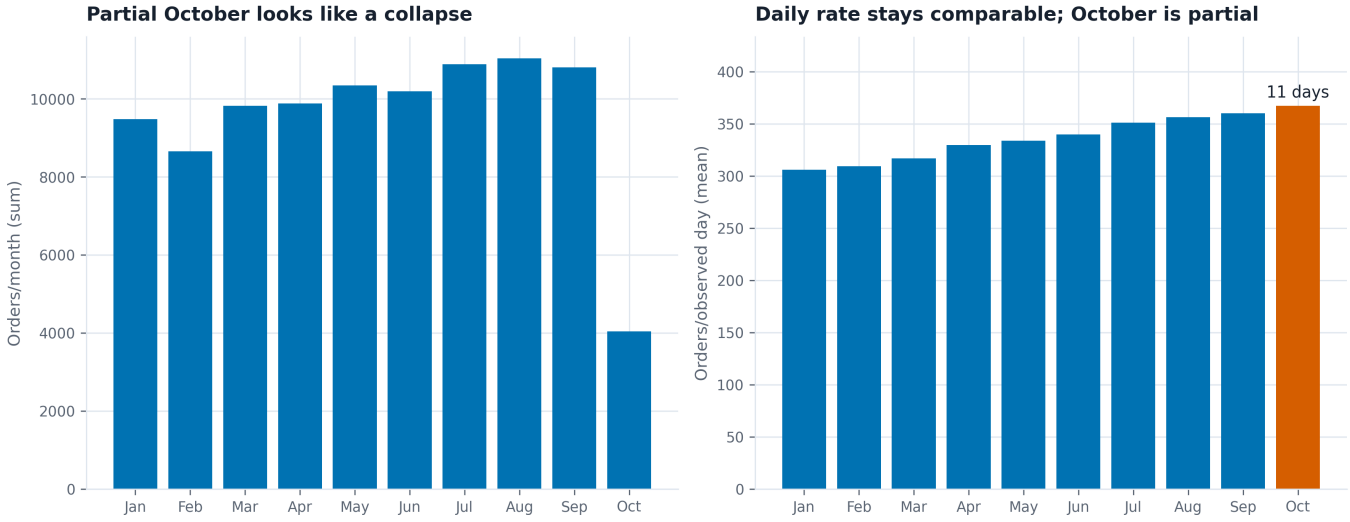
Mechanism

Selected seven points fall 158.8 to 92.2 (-41.9%). Same-month 2025 vs 2024 changes range 5.9% to 13.3%.

Redesign

Show all 36 months and compare matching months across years; a seasonal downswing is not a long-run decline.

Case 6: Unequal exposure in a partial month



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

Orders appear to collapse in October.

Decoy

Monthly sum is correct for a flow and the bars start at zero; the periods are not equally complete.

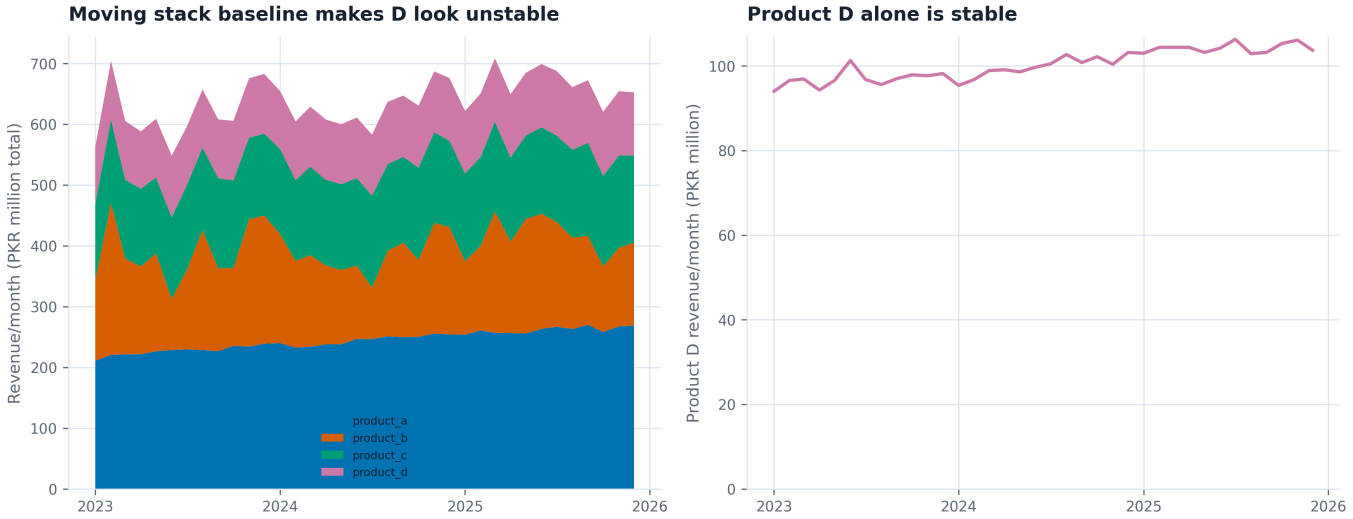
Mechanism

September 10805 in 30 days (360.2/day); October 4041 in 11 days (367.4/day), extracted 2025-10-11.

Redesign

Retain zero baseline, label partial exposure, and compare per-observed-day rates instead of unequal monthly totals.

Case 7: Volatility transferred through a stack baseline



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

The upper envelope seems to show erratic product D.

Decoy

The stack is legal: products are exclusive, exhaustive and positive; the issue is comparison, not arithmetic.

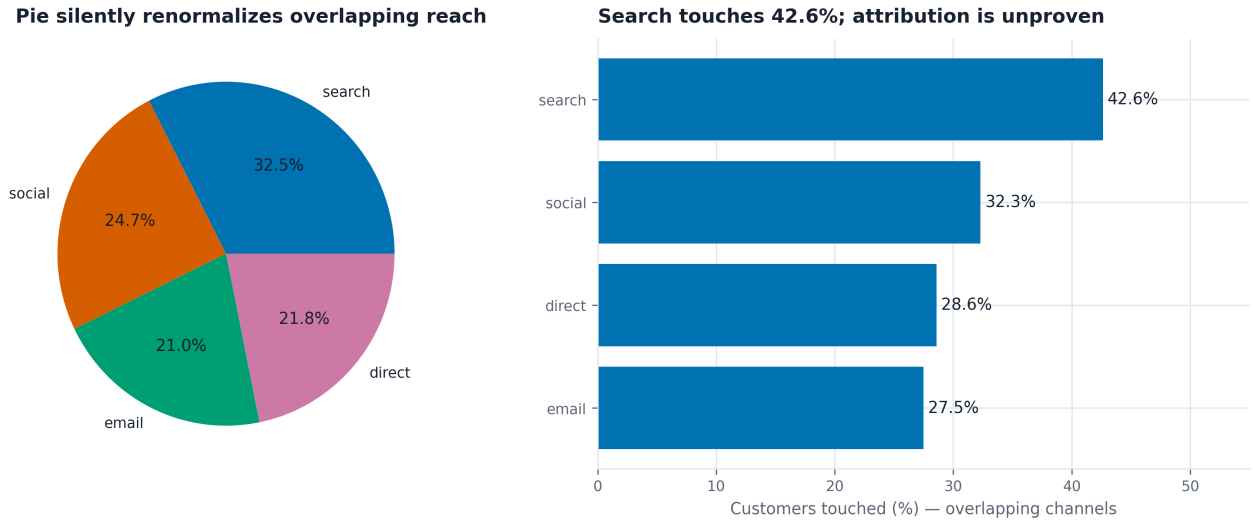
Mechanism

Total envelope 548.3-708.7; D range 94.0-106.3, SD 3.55; lower three baseline SD 39.27.

Redesign

Keep the legal composition stack and add an unstacked D series beside it. Band thickness, not top boundary, is D.

Case 8: Non-exclusive denominator



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

Search appears to cause 43% of customers.

Decoy

Each reach percentage is arithmetically correct with 4,000 customers as denominator; neither rounding nor a bad axis explains the sum.

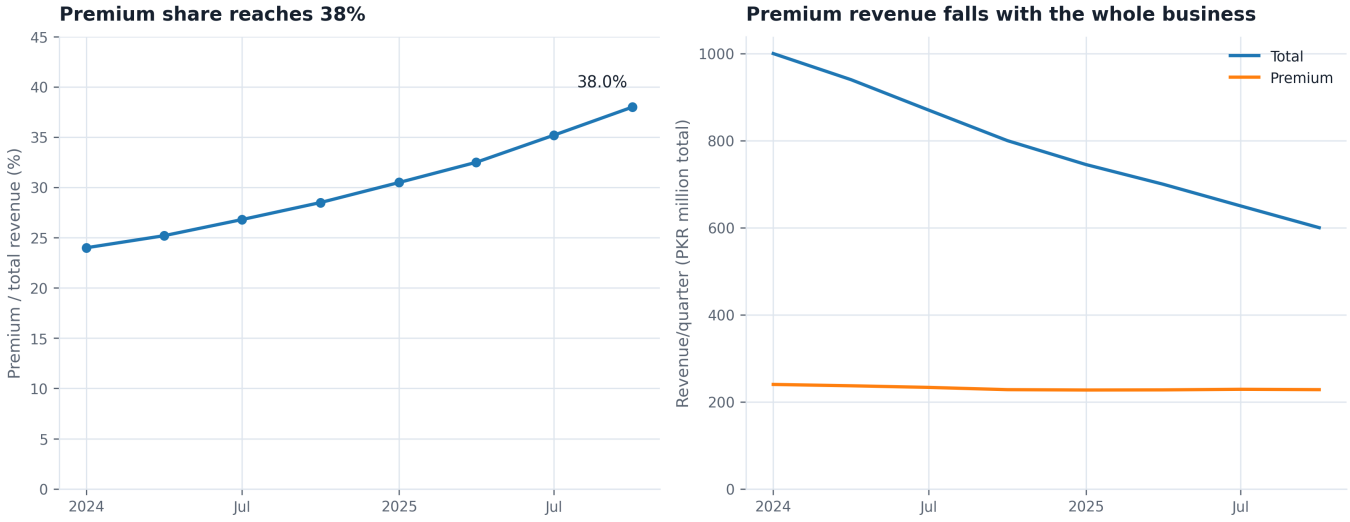
Mechanism

Shares {'touch_search': 42.62, 'touch_social': 32.3, 'touch_email': 27.48, 'touch_direct': 28.58} sum to 131.0%; 1105 customers have multiple touches. Pie area for search is 32.54%, not 42.6%.

Redesign

Separate reach bars use a common customer denominator; labels explicitly say overlap, and touching a channel is not causal attribution.

Case 9: Rising fraction of a shrinking denominator



Synthetic course data: CDB601220, Superior University Lahore, Week 4. 12×4.8 in. Left: analyst recipe; right: honest redesign. Intentionally invalid recipe shown for diagnosis only.

Conclusion

A record share suggests premium is growing.

Decoy

The 24% to 38% share is true, exclusive and meaningful for mix; it is incomplete for performance.

Mechanism

Total 1000.0 to 600.0; premium 240.0 to 228.0, a -5.0% change. Denominator shrinks faster.

Redesign

Keep the share chart and add absolute premium and total revenue on a separate shared-unit panel, never a dual axis.